



PATNet: Patch-to-pixel attention-aware transformer network for RGB-D and RGB-T salient object detection

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ABSTRACT

Multimodal salient object detection (SOD) combines different modal images to generate the most visually appealing saliency map. When fusing multimodal and multiscale features, maintaining the integrity and fine granularity of the target is critical for improving the performance of multimodal SOD. The fine-grained information differences between the modalities and the size of the features in the transformer prevent most existing studies from guaranteeing both granularities. Therefore, we propose a patch-to-pixel attention-aware transformer network (PATNet) to overcome these problems, whereby the integrity and fine-grained details of the saliency map are preserved by employing a decision-transformation strategy to map global patches onto local pixels. Specifically, PATNet consists of the shared attention fusion module (SAFM), adjacent modeling fusion module (AMFM), and fine-grained mapping module (FMM). SAFM enhances the consistency between multimodal features through a shared attention matrix and an identical convolutional feed-forward network. Meanwhile, AMFM enhances low-resolution features by modeling neighboring features to avoid the aliasing effect of upsampling. In the output stage, FMM is responsible for mapping the feature maps represented by patches onto pixels and restoring the salient object details. Numerous experimental results demonstrate that PATNet outperforms 24 state-of-the-art methods on six RGB-D and three RGB-T datasets. The source code is publicly available at <https://github.com/LitterMa-820/PATNet>.

1. Introduction

In the field of computer vision, salient object detection (SOD) is a popular research topic that aims to extract the most visually appealing part of a given image. SOD has a wide range of applications in various fields including object tracking [1,2], image captioning [3], image smoothing [4], image segmentation [5,6], and image compression [7]. The different modal images introduced in SOD enhance the performance in various application scenarios. By introducing depth information, RGB-D SOD can obtain richer structural information, allowing salient targets to be better separated from the image. Similarly, RGB-T SOD can operate in extreme environments with the assistance of thermal images, which are insensitive to illumination conditions.

In recent years, vision transformer-based multimodal SOD methods [8–14] have surpassed convolution-based methods [15–22] and reached important milestones. Although existing multimodal SOD methods achieve good results, they are insufficient for the fine-grained segmentation of saliency maps. As depicted in Fig. 1, both state-of-the-art methods exhibit a noticeable loss of detail at the edges of the flowers and legs of the insects, which can be attributed to the following

shortcomings of the existing methods, detailed in Fig. 2. First, the depth images are inconsistent with the fine-grained information in the RGB images. Second, the patches obtained by patch embedding make it difficult to express the details in saliency maps. Third, excessive reliance on bilinear interpolation in the decoding and output processes results in the accumulation of aliasing effects. Given the abovementioned issues, existing methods perform unsatisfactorily regarding the details of saliency maps. To address these issues, we propose PATNet to generate complete and fine-grained saliency maps by enhancing modal consistency and reducing the loss of fine-grained information in the decoder. PATNet employs two pyramid pooling transformers (P2T) with multiscale pooling as a multimodal encoder to extract the features expressed by patches. The decoding process is divided into two stages consisting of patch and pixel-levels. After processing the features at different stages, supervised learning is performed at different granularities, enabling the model to jointly decide the salient parts of the image at the patch and pixel levels, thereby obtaining complete and fine-grained saliency maps.

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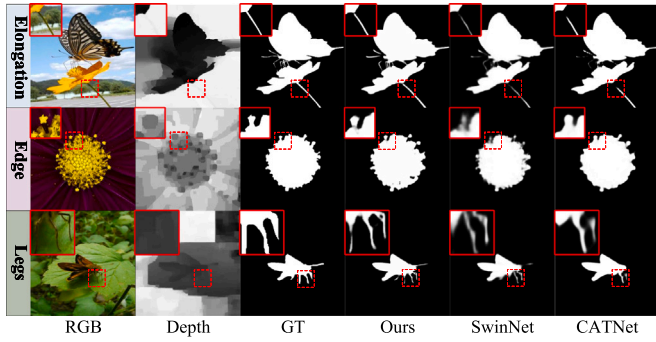


Fig. 1. Visual comparison of our approach with two state-of-the-art methods on several fine-grained examples. The transformer-based SwinNet and CATNet with an input size of 384×384 exhibit poor performance in the detailed parts.

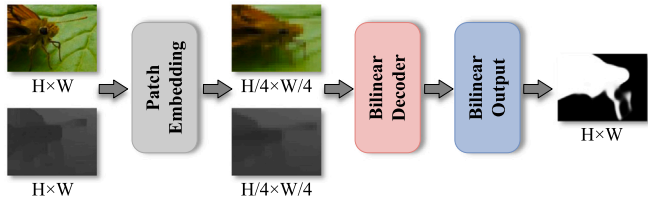


Fig. 2. Shortcomings of existing transformer-based methods. The patch embedding in the encoding phase leads to a loss of detail, and the bilinear interpolation during decoding and output leads to aliasing effects.

The patch-level stage primarily comprises two innovative multi-head attention modules, namely the shared attention fusion module (SAFM) and adjacent modeling fusion module (AMFM). In SAFM, the jointly determined multimodal features in the shared attention matrix achieve complementarity, and an identical feedforward block is used to explore modal coherence. AMFM uses an adjacent high-resolution feature map as a query to reconstruct a low-resolution feature map with double matrix multiplication to avoid the aliasing effect caused by bilinear interpolation. The features of different modalities and scales are processed using SAFM and AMFM to assist the model in making global saliency decisions through supervised learning. The feature map expressed by patches can accurately output the central part of the salient map, but its size is only one-sixteenth of the original image, which is unsuitable for fine-grained SOD. In the transformer architecture, patch embedding compresses image pixels into channel dimensions to reduce the cost of the attention mechanism. The patch embedding process in the transformer can be reversed by mapping patches to pixels, thereby improving the detailed performance of saliency maps. Therefore, a fine-grained mapping module (FMM) was designed to weigh and map the patches onto pixels at the pixel-level stage. FMM allows the upper-level global saliency decisions to be transformed into fine-grained outputs of local decisions. As shown in the third column of Fig. 1, the saliency maps obtained after the two decoding stages described above provide better details than those obtained by other same-scale input methods.

The main contributions can be summarized as follows:

- The proposed PATNet first uses P2T as a multimodal feature encoder and gradually obtains saliency maps through two stages of different granularity processing in the decoder. Using PATNet, state-of-the-art results on six common RGB-D and three RGB-T datasets were obtained in multimodal salient object detection.
- SAFM is proposed to achieve complementarity and consistency in multimodal feature fusion through a shared attention matrix and convolutional feed-forward network.
- AMFM is proposed to fuse features at different scales, whereby low-resolution feature maps are reconstructed into

high-resolution feature maps by calculating the correlation matrix from feature maps of different scales.

- FMM is proposed to help the saliency maps reduce blurring and enhance local details by establishing a mapping relationship between patches and pixels.

2. Related work

2.1. RGB salient object detection

The task of RGB salient object detection is to extract the most visually attractive regions from a single RGB image. To get a saliency map with more precise edges, Wei et al. [23] decoupled the ground truth into detail maps and body maps and used them to supervise the model training, leading to impressive performance in RGB saliency detection. Wei et al. [16] also proposed a cascade feedback decoder for the SOD and introduced the PPA loss to assign more weights to edge pixels. To improve the accuracy of saliency information, Wu et al. [20] proposed the EDN to further compress the deep semantic information to get stronger saliency detection capability. Liu et al. [24,25] proposed to use multiple pooling layers to enhance saliency semantic information. Integrity learning was proposed by Zhuge et al. [26] to align the integrity of micro and macro saliency. In recent years, cascading and iteration have also become commonly used methods for detecting salient objects. Fang et al. [22] proposed the LC³Net to better aggregate adjacent and non-adjacent semantic information using contextual cascading feedback. Wang et al. [27] proposed the MENet, which employs the boundary sensitivity, content integrity, iterative refinement, and frequency decomposition mechanisms of the human visual system. Various attention-based methods are also widely used in SOD. Li et al. [28] proposed to densely aggregate multiple attention units, enabling deep feature maps to receive attention information from shallow feature maps. Lee et al. [29] proposed the TRACER to detect salient objects with well-defined edges by merging attention-guided tracking modules.

So far, unimodal SOD has progressed considerably. However, introducing multimodal information is necessary to handle varied application scenarios and achieve better segmentation results.

2.2. RGB-D salient object detection

RGB-D salient object detection aims to enhance saliency information in RGB images by incorporating depth images. Depth images depict 3D information about the scene and aid in distinguishing the foreground from the background. Convolution-based methods have played an essential role in the development of RGB-D SOD. Li et al. proposed the ICNet [30] and CMWNet [31], where the former employs an information conversion module to interactively and adaptively fuse high-level RGB and depth features, and the latter utilizes three RGB and depth interaction modules to fuse multimodal multiscale features from low, medium, and high levels. Bi et al. [32] implemented cross-modal feature fusion in the encoding stage by feeding intermediate features back to the encoder. Zhang et al. [33] introduced a criss-cross dynamic filter network in the C2DFNet to achieve dynamic selection between two modalities. Chen et al. [15] proposed the RD3D, which utilizes 3D convolution for deep multimodal feature fusion. Zhao et al. [34] achieved an approximation of ImageNet pre-training by detour maps for pre-training of the SSLSOD. Recently, the vision transformer [35] architecture and Canny [36] edge detection have been applied to RGB-D SOD. Liu et al. [9] developed the SwinNet using the Swin Transformer [37] as the backbone and used depth features to generate edge maps as an aid. Similarly, Sun et al. [14] proposed the CATNet using Swin Transformer, cascade structure, and edge map to generate saliency maps. Chen et al. [38] used the PVT [39] as a backbone, fused the multimodal and multiscale features using the MTF and MTE, respectively, and extracted edge information by the SOEE. Asymmetric

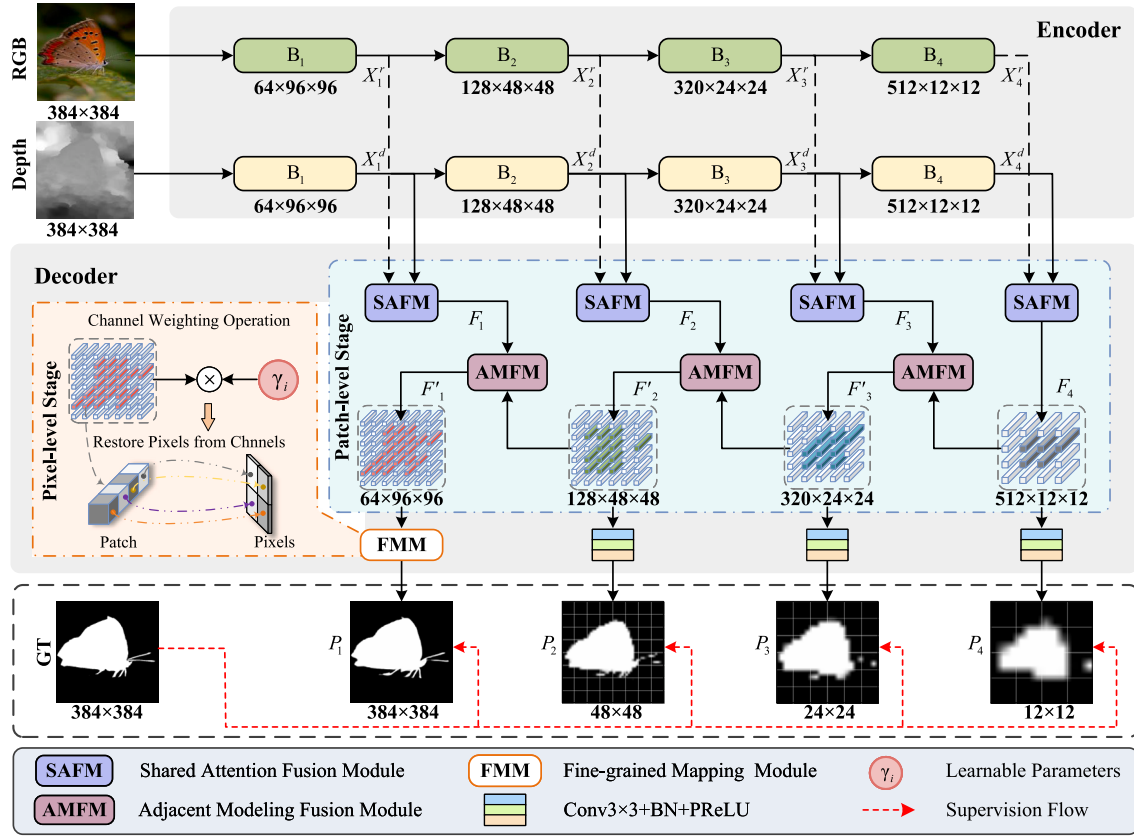


Fig. 3. The overview of PATNet architecture. Two separate P2T-base encoders are used to encode the two modal inputs. Four SAFMs and three AMFMs fuse features from different modalities and scales at the patch-level stage. Then, FMM maps patches of features to pixels at the pixel-level stage. Finally, three coarse-grained and one fine-grained ground truth are used for supervised learning on the four output branches.

encoder structures have been recently developed, with Liu et al. [10] using the HRFormer [40] to extract RGB features and the ResNet to extract auxiliary modal features. Hussain et al. [41] used the DPT [42] as an encoder and a monocular depth estimation network to obtain depth maps for RGB-D SOD. Some researchers also used variants of the self-attention module to fuse cross-modal and cross-scale features in the decoder. Liu et al. [12] introduced a shared-weight triplet transformer module in the decoder stage to establish long-range dependencies across layers. Pang et al. [11] modified the self-attention module to the CMU for the decoding stage.

Although these methods have achieved acceptable results, the imaging results are often accompanied by incomplete segmentation results, blurred edges, and loss of local details. The addition of edge detection techniques in RGB-D SOD did not improve the problems and increased the cost of supervision learning.

2.3. RGB-T salient object detection

RGB-T SOD refers to using a thermal image along with an RGB image to detect salient objects under challenging conditions, including low light and extreme weather. RGB-T SOD was developed later, and the technology usually follows RGB-D SOD. Tu et al. [43] developed a dual-decoder structure that integrates global, multimodal, and multiscale features, achieving promising results in both RGB-T and RGB-D SOD tasks. Recent studies indicate that erroneous thermal radiation in thermal images may affect saliency maps and that thermal maps are unnecessary for all situations. Cong et al. [44] proposed to decide whether thermal features are necessary using a global illumination estimation module. Liao et al. [45] attempted to enable RGB and thermal features to complement each other adaptively through early fusion in the encoding stage. Some researchers have used knowledge distillation and

data enhancement to improve the capabilities of RGB-T SOD networks. Zhou et al. [46] improved performance for the Wavelet by knowledge distillation. Liu et al. [47] implemented scribble supervision on the model by constructing scribble annotations datasets.

While these related works reduce the interference of thermal images, improving the semantic understanding of the network is fundamental to solving the problem. Moreover, generating detailed saliency maps under thermal interference remains a challenge that has yet to be solved. Our proposed PATNet network has powerful semantic processing capabilities to accurately find salient objects in multimodal SOD tasks and generate fine-grained saliency maps without the additional supervision of edge maps.

3. Proposed method

3.1. Overview

The structure of our proposed model, which uses the RGB-D multimodal task as an example, is illustrated in Fig. 3. The encoder of PATNet consists of two P2T-base backbones, and the extracted features are categorized into four levels B_i ($i = 1, \dots, 4$). In addition, the different modal features for each level are denoted as X_i^r and X_i^d , respectively. Note that X_i^r and X_i^d have the same resolution and channel dimension. At the patch-level stage, the two modal outputs X_i^r and X_i^d for each level are fused across modes using SAFM to obtain the fused feature F_i . After cross-modal fusion, $F_{1 \sim 3}$ are combined with the low-resolution features from the previous layer for cross-scale feature fusion by AMFM to obtain $F'_{1 \sim 3}$. At the pixel-level stage, FMM maps and outputs P_1 for fine-grained supervision using full-size labels, while a set of CBPs outputs $P_{2 \sim 4}$ for coarse-grained supervision using downsampled labels.

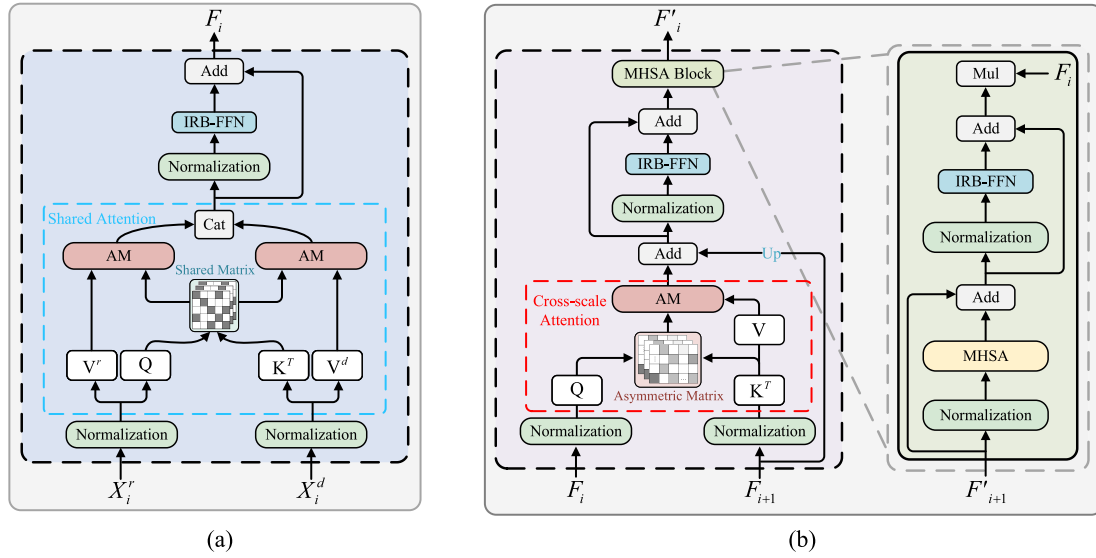


Fig. 4. Two novel multi-head attention modules in the proposed network. MHSA denotes multi-head self-attention. AM denotes a series of operations after obtaining the attention matrix, including Softmax and matrix multiplication. (a) Shared Attention Fusion Module. (b) Adjacent Modeling Fusion Module.

3.2. Shared attention fusion module

We designed a two-branch multi-head shared attention fusion module at the patch-level stage to enhance inter-modal consistency and complementarity. The features X_i^r and X_i^d from two modalities are multiplied to obtain a shared attention matrix. This matrix retains the consistency information of the two modalities and is passed to the shared attention operation of the two branches to achieve modal complementarity. Besides, an inverse residual block feed-forward network (IRB-FFN) combines the information from both branches to make the intermodal information consistent. The structure of SAFM is shown in Fig. 4(a).

Specifically, given features X_i^r and X_i^d , we first perform a Layer-Norm and linear projection, which can be expressed as:

$$\begin{aligned} Q, V^r &= \psi(LN(X_i^r)) \\ K, V^d &= \psi(LN(X_i^d)) \end{aligned} \quad (1)$$

where X_i^r represents the RGB feature map, X_i^d represents the depth or thermal feature map, $LN(\cdot)$ represents the LayerNorm operation, and $\psi(\cdot)$ denotes the linear projection operation. Q , V^r , K , and V^d are variables used in the shared attention operation.

Then, we perform a matrix multiplication between Q and K to obtain the shared attention matrix M_s for the RGB modality branch. Since $AB^T = (BA^T)^T$ in matrix operations, the attention matrix M_s^T for the depth modality can be obtained by sharing M_s and transposing it. Note that the Softmax operation is not applied at this stage. The above procedure can be expressed as:

$$M_s = \frac{QK^T}{\sqrt{D/N}} \quad (2)$$

where D represents the length of the features dimension, and N represents the number of heads in the multi-head operation.

After obtaining the shared matrix M_s , the remaining attention manipulations of the double branch are represented as follows:

$$\begin{aligned} X_i^{r'} &= \delta(M_s)V^r + X_i^r \\ X_i^{d'} &= \delta(M_s^T)V^d + X_i^d \end{aligned} \quad (3)$$

where $\delta(\cdot)$ represents the Softmax activation function. $X_i^{r'}$ and $X_i^{d'}$ denote the results of each branch in shared attention, respectively.

In the feed-forward network, we concatenate the features of the two modalities in the channel dimension and fuse them with an IRB-FFN with an expansion ratio of 2. At last, a convolution layer with a

3×3 sized kernel halves the number of feature channels. The above procedure can be represented as:

$$\begin{aligned} fuse &= Cat(X_i^{r'}, X_i^{d'}) \\ fuse' &= IRB(fuse) + fuse \\ F_i &= Conv_{3 \times 3}(fuse') \end{aligned} \quad (4)$$

where $Cat(\cdot)$ is the concatenation operation, $Conv_{n \times n}(\cdot)$ represents a convolution layer with a kernel size of $n \times n$, and F_i represents the output of this module.

In addition, the IRB in Eq. (4) is expressed as follows:

$$\begin{cases} x_1 = Conv_{1 \times 1}(x) \\ x_2 = DWConv_{3 \times 3}(x_1) \\ x_3 = Conv_{1 \times 1}(x_2) \end{cases} \quad (5)$$

where $DWConv_{3 \times 3}(\cdot)$ means the depthwise convolution with a kernel size of 3×3 .

3.3. Adjacent modeling fusion module

AMFM is responsible for upsampling and fusing features from two adjacent scales at the patch-level stage. To minimize the aliasing effect during the upsampling process, we use high-resolution features as Q and low-resolution features as K, V for cross-scale attention. In this process, the low-resolution will be adaptively reconstructed into high-resolution features by the asymmetric attention matrix. Then, the multiscale features are fused by multi-head self-attention and dot product. AMFM has three advantages over these methods [11,13,14]. First, the patches newly generated in the high-resolution features reconstructed by AMFM come from the contributions of all patches. Second, the absence of bilinear interpolation before cross-scale attention avoids the generation of meaningless channels. Third, fewer patches reduce the computational cost of matrix multiplication. The structure of AMFM is shown in Fig. 4(b).

To start with, we apply LayerNorm and linear projection to project F_i ($i = 1, 2, 3$) as Q and F_{i+1} as K, V . Meanwhile, the residual is obtained by upsampling the low-resolution features via bilinear interpolation, which guarantees the module's lower bound. These operations are expressed as follows:

$$\begin{aligned} Q &= \psi(LN(F_{i+1})) \\ K, V &= \psi(LN(F_i)) \\ R &= Bi(LN(F_{i+1})) \end{aligned} \quad (6)$$

where F_i represents the high-resolution feature map of the current level, F_{i+1} represents the low-resolution feature map output from the previous level, and $Bi(\cdot)$ represents the bilinear interpolation.

Next, Q and K are matrix-multiplied to obtain an asymmetric matrix M of correspondences between patches of the two resolution features. Moreover, the low-resolution features are upsampled to the high-resolution feature scale after two matrix multiplications. The feed-forward network here also uses IRB-FFN. The process can be represented as follows:

$$\begin{aligned} M &= \delta\left(\frac{QK^T}{\sqrt{D/N}}\right) \\ \overline{F_{i+1}} &= MV + R \\ F_{i+1}^h &= IRB(\overline{F_{i+1}}) + \overline{F_{i+1}} \end{aligned} \quad (7)$$

where Q, V are matrices of size $n_1 \times D$ and K^T is a matrix of size $D \times n_2$. After QK^T , the asymmetric correlation matrix of size $n_1 \times n_2$ is obtained. The upsampling process is $M \times V$ to obtain high-resolution features with resolution $n_1 \times D$.

Then, we use a multi-head self-attention module to enhance the multiscale and multimodal information in the reconstructed features, as follows:

$$\begin{aligned} Q, K, V &= \psi(LN(F_{i+1}^h)) \\ \overline{F_{i+1}^h} &= \delta\left(\frac{QK^T}{\sqrt{D/N}}\right)V + F_{i+1}^h \\ F_{i+1}^{h'} &= IRB(\overline{F_{i+1}^h}) + \overline{F_{i+1}^h} \end{aligned} \quad (8)$$

where $F_{i+1}^{h'}$ is the obtained feature after enhancement.

Finally, $F_{i+1}^{h'}$ is fused with F_i by a Hadamard product and the CBP ($Conv_{3 \times 3} + BatchNorm + PReLU$), which can be formulated as:

$$F_i' = CBP(F_i \cdot F_{i+1}^{h'}) \quad (9)$$

where F_i' denotes the output of AMFM.

The saliency information contained in F_i' after processing the patch-level operation in SAFM and AMFM is gradually enhanced and aids the model's decision-making at different scales in supervised learning.

3.4. Fine-grained mapping module

After the features are processed at the patch-level stage, they have a powerful coarse-grained saliency expression. However, these features must also be processed at the pixel-level stage to obtain saliency maps with better confidence and richer details. As illustrated in Fig. 3, FMM establishes a mapping relationship between pixels from the input image to the output image. There are two steps involved in this process. In the first step, the channels of the feature are reduced and weighted simultaneously. In the second step, the elements in the channels are mapped to pixels in the final saliency map. Compared to bilinear interpolation on patches sampled at the input scale, FMM suppresses the aliasing effect more effectively and produces saliency maps with more details.

Specifically, the feature map adjusts the channel weights by 64 learnable parameters γ . Meanwhile, we adjust the number of channels of F_i' to the required upsampling multiplier r by CBP, as follows:

$$\begin{aligned} F_1^{w'} &= F_1' \cdot \{\gamma_1, \gamma_2, \gamma_3, \dots, \gamma_j\} \\ F_1^{w''} &= CBP(F_1^{w'}) \end{aligned} \quad (10)$$

where $j = 64$, it means that the F_1' have 64 channels.

Then, channel mapping converts the 16-channel patch features into a one-dimensional saliency map. In the mapping operation, pixel shuffling [48] allows adjacent pixels to be from the same channel and reduces the number of channels in the feature map, which helps generate saliency maps. In addition, the previous channel weighting operation can mask some incorrect channels to avoid artifacts. Based

on the above discussion, the fine-grained output can be obtained from the following equation:

$$P_1 = PS(F_1^{w''}) \quad (11)$$

where P_1 is the pixel-level prediction result of PATNet and $PS(\cdot)$ represents pixel shuffling operation.

Besides, we use a set of CBPs to reduce channel dimensions of $F_{2 \sim 3}'$ and F_4 to get three coarse-grained outputs, which are used for patch-level global supervised learning, as follows:

$$\begin{aligned} P_{2 \sim 3} &= CBP(F_{2 \sim 3}') \\ P_4 &= CBP(F_4) \end{aligned} \quad (12)$$

where $P_{2 \sim 4}$ denote the patch-level saliency maps.

3.5. Micro design

Establishing a multi-head mechanism relation between the encoder and decoder in patch-level operations could improve the model's efficiency. To optimize this, we have adopted the same multi-head grouping as the encoder for SAFM and AMFM in stage $B_{1 \sim 4}$, with the specific number of multi-heads $\{1, 2, 5, 8\}$.

In AMFM, we use the Hadamard intersection product to fuse the F_i and $F_{i+1}^{h'}$, which extracts the saliency map more efficiently than summation.

In the CBP, we utilize PReLU as the activation function instead of the traditional ReLU. PReLU improves the mapping ability of pixel values in the negative direction and the expression of features, resulting in a more apparent distinction between background and foreground in the generated image.

3.6. Loss function

We use the same loss function as F³Net [16] to supervise all branch outputs during decoding, which include weighted binary cross-entropy and weighted intersection over union, and it is defined as:

$$\mathcal{L}_i = \mathcal{L}_{bce}^w(GT_i, P_i) + \mathcal{L}_{iou}^w(GT_i, P_i) \quad (13)$$

where $\mathcal{L}_i$ denotes the loss between the prediction and ground truth at each layer, $\mathcal{L}_{bce}^w$ and $\mathcal{L}_{iou}^w$ represent the weighted binary cross-entropy and weighted intersection over union losses, respectively. GT_i is the ground truth corresponding to the resolution of P_i .

We use the sum of $\mathcal{L}_i$ for each layer as the total loss during training, expressed as follows:

$$\mathcal{L}_{sod} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4 \quad (14)$$

where $\mathcal{L}_{sod}$ denotes the sum of the four granularity branching losses.

4. Experiments

4.1. Experimental setup

RGB-D Datasets. DUT-RGBD [49] contains 800 images in the training set and 400 in the testing set. NJUD [50] contains 1985 images collected from 3D movies and the internet. NLPR [51] consists of 1000 RGB-D image pairs collected using Microsoft Kinect. DES [52] contains 135 indoor images. STEREO [53] consists of 1000 image pairs from real and virtual scenes. SIP [54] contains 929 high-resolution portrait image pairs. Following these works [11,34,49,55–57], that is 1485 samples from NJUD, 700 from NLPR and 800 from DUT-RGBD as the training set.

RGB-T Datasets. VT-821 [58] contains 821 pairs of manually annotated RGB-T images. VT-1000 [59] includes 1000 pairs of relatively simple and aligned scenes. VT-5000 [60] consists of 5000 pairs of images with complex scenes and different objects. According to [43], we select 2500 image pairs from VT5000 as the training set and the rest of the image pairs along with VT821 and VT1000 as the test set.

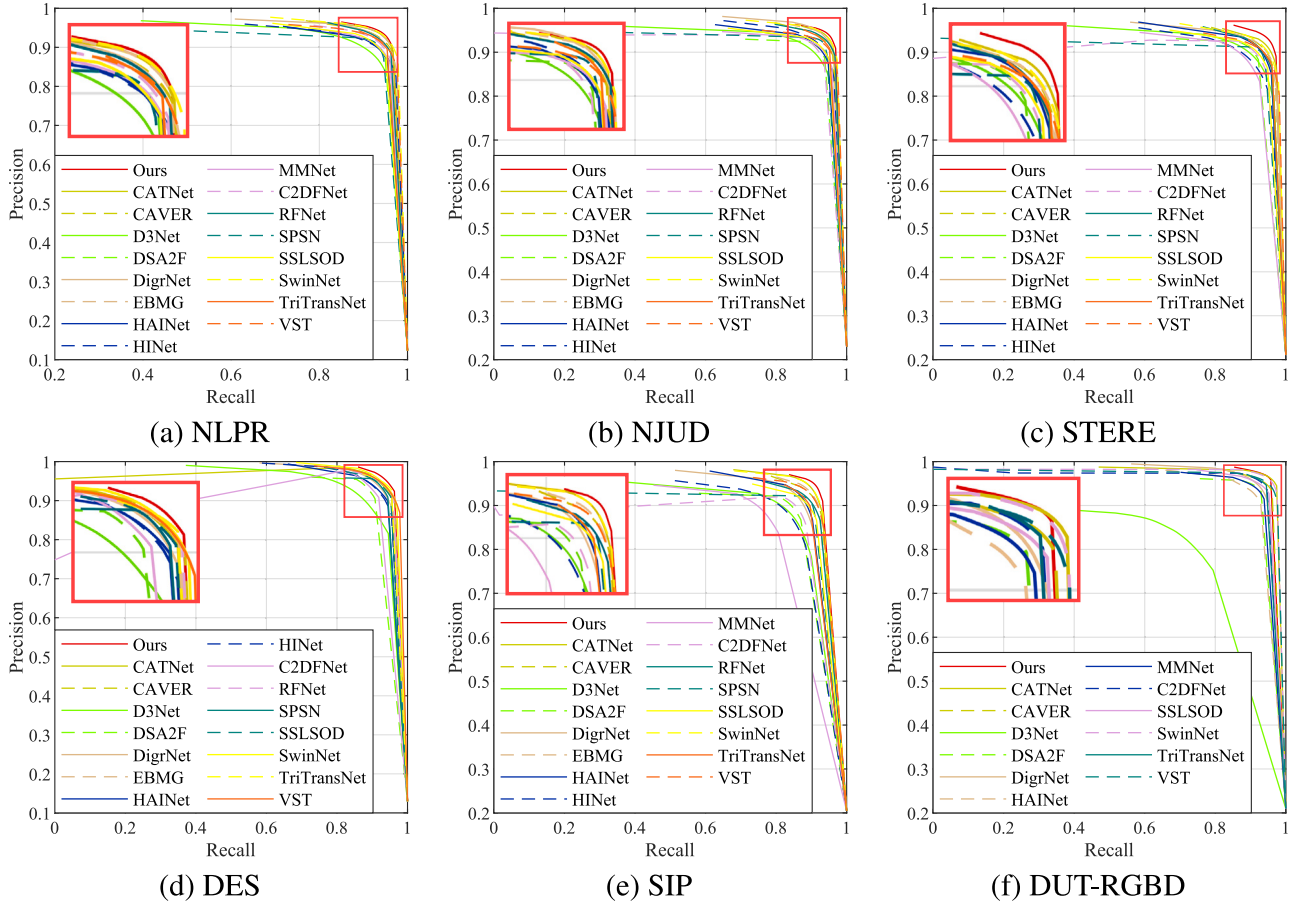


Fig. 5. PR curves comparison of different models on six RGB-D datasets. The solid red line represents our PATNet.

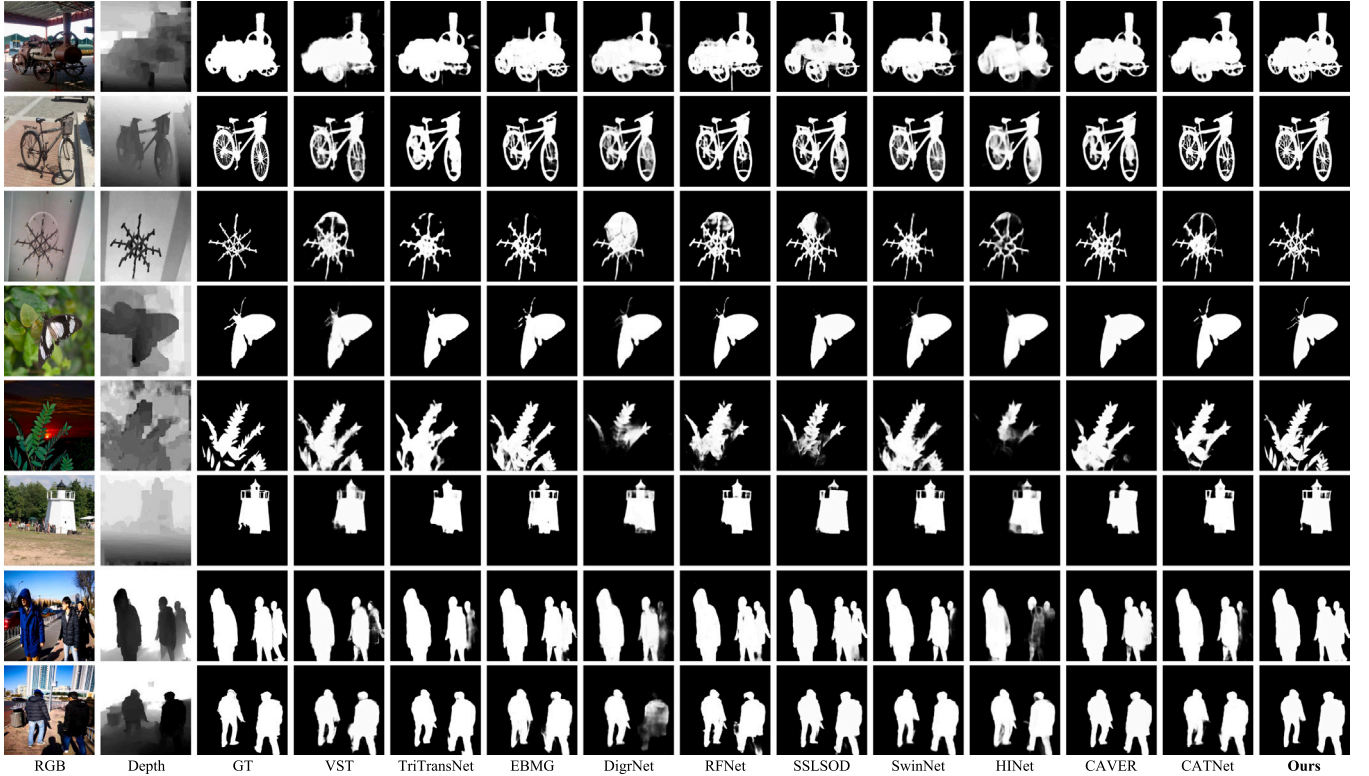


Fig. 6. Visual comparison of our model with other SOTA RGB-D models.

Table 1
S-measure, max F-measure, adaptive E-measure, MAE comparisons with different RGB-D models. the top scores are indicated in red, the second-best scores are indicated in green, and the third-best scores are indicated in blue.

Datasets	Metric	D3Net TNLS20	HAINet TIP21	MMNet TCSVT22	DSAF2F CVPR21	VST ICCV21	TrifTransNet MM21	EBMG NIPS21	SPSN ECCV22	SSLSOD AAAI22	C2DFNet TMM22	RFNet 3DV22	DigrNet TMM22	SwinNet TCSVT22	HINet PR23	CAVER TIP23	CATNet TMM23	PATNet Ours
NUPR	S1	.912	.922	.925	.918	.931	.928	.938	.923	.922	.927	.931	.935	.941	.922	.934	.940	.942
	F ₁ ↑	.907	.915	.918	.916	.927	.929	.940	.918	.922	.926	.933	.934	.941	.915	.934	.939	.943
	E _s ↓	.945	.950	.955	.952	.956	.964	.969	.957	.960	.960	.961	.957	.969	.950	.966	.971	.971
	MA ↓	.030	.026	.023	.024	.023	.020	.018	.023	.025	.021	.020	.023	.018	.026	.021	.018	.016
NIUD	S1	.900	.915	.910	.904	.922	.920	.929	.918	.909	.908	.926	.933	.935	.915	.925	.932	.936
	F ₁ ↑	.910	.925	.918	.917	.926	.934	.941	.927	.922	.918	.937	.943	.943	.925	.935	.943	.946
	E _s ↓	.916	.939	.938	.937	.943	.954	.955	.934	.939	.941	.952	.954	.954	.939	.953	.958	.960
	MA ↓	.046	.039	.039	.039	.034	.030	.028	.033	.038	.039	.030	.028	.027	.039	.029	.026	.025
STERE	S1	.899	.892	.884	.897	.913	.908	.916	.906	.904	.902	.911	.916	.919	.892	.917	.921	.932
	F ₁ ↑	.904	.897	.896	.910	.915	.919	.923	.910	.912	.911	.921	.921	.926	.897	.923	.930	.940
	E _s ↓	.924	.927	.931	.942	.942	.950	.952	.931	.939	.942	.946	.943	.950	.927	.951	.954	.962
	MA ↓	.046	.049	.046	.039	.038	.033	.032	.036	.039	.038	.036	.038	.033	.049	.032	.030	.025
DES	S1	.898	.927	-	.916	.943	.943	.938	.937	.936	.921	.942	.937	.945	.927	.942	.941	.948
	F ₁ ↑	.909	.937	-	.930	.947	.946	.947	.942	.944	.937	.947	.943	.952	.937	.951	.947	.957
	E _s ↓	.952	.969	-	.955	.979	.981	.977	.975	.978	.964	.982	.970	.980	.969	.979	.978	.983
	MA ↓	.031	.022	-	.023	.017	.014	.016	.016	.017	.020	.015	.020	.016	.022	.016	.017	.013
SIP	S1	.860	.856	.824	.862	.904	.886	.902	.892	.888	.871	.894	.885	.911	.856	.904	.911	.917
	F ₁ ↑	.881	.880	.861	.891	.925	.916	.929	.910	.908	.895	.916	.913	.936	.880	.925	.937	.942
	E _s ↓	.902	.899	.871	.911	.941	.929	.942	.932	.927	.918	.932	.919	.947	.899	.944	.948	.953
	MA ↓	.063	.066	.080	.057	.040	.043	.037	.043	.046	.053	.043	.053	.035	.066	.037	.034	.031
DUT-RGBD	S1	.775	.909	.920	.921	.942	.932	-	-	.929	.933	-	.926	.948	-	.937	.953	.949
	F ₁ ↑	.797	.933	.938	.938	.954	.951	-	-	.946	.951	-	.948	.962	-	.951	.963	.964
	E _s ↓	.849	.934	.954	.956	.960	.966	-	-	.958	.963	-	.951	.973	-	.967	.974	.974
	MA ↓	.097	.038	.033	.031	.025	.025	-	-	.029	.026	-	.033	.020	-	.026	.020	.019
Params(M) FLOPs(G)		-	59.8	-	36.5	83.0	138.7	-	37.0	74.2	47.5	129.8	166.7	198.8	98.9	93.8	262.6	95.9
		-	181.4	-	91.4	31.0	292.3	-	501.7	144.0	11.1	112.5	52.7	124.7	389.7	63.9	341.8	51.1

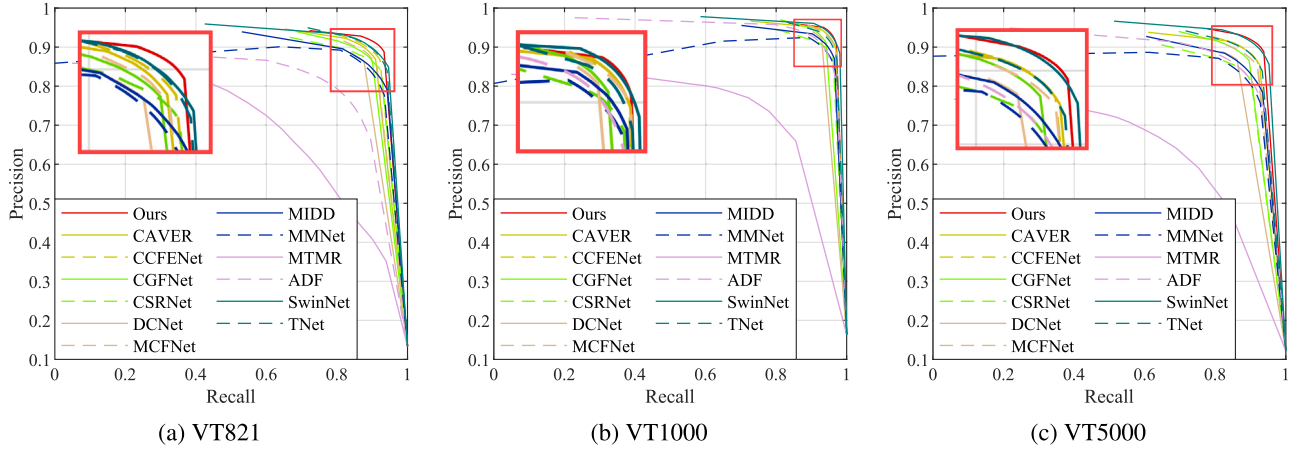


Fig. 7. PR curves comparison of different models on three RGB-T datasets. The solid red line represents our PATNet.

Table 2

S-measure, max F-measure, adaptive E-measure, MAE comparisons with different RGB-T models. the top scores are indicated in red, the second-best scores are indicated in green, and the third-best scores are indicated in blue.

Datasets	Metric	MTMR IGTA18	MIDD TIP21	ADF TMM22	CSRNet TCSVT22	MMNet TCSVT22	CGFNet TCSVT22	DCNet TIP22	TNet TMM22	CCFNet TCSVT22	SwinNet TCSVT22	CAVER TIP23	PATNet Ours
VT821	S $\uparrow$	.652	.871	.810	.884	.873	.880	.877	.898	.899	.903	.897	.914
	F β $\uparrow$	.691	.874	.804	.880	.866	.885	.868	.904	.901	.903	.897	.914
	E $\downarrow$ $\uparrow$	.786	.898	.845	.912	.894	.918	.920	.924	.934	.928	.934	.938
	M $\downarrow$	.190	.045	.077	.038	.040	.038	.033	.030	.027	.030	.026	.024
VT1000	S $\uparrow$	.705	.915	.909	.918	.914	.922	.922	.928	.933	.938	.937	.941
	F β $\uparrow$	.755	.926	.923	.918	.915	.936	.928	.937	.941	.948	.945	.948
	E $\downarrow$ $\uparrow$	.836	.942	.922	.940	.932	.955	.961	.953	.963	.957	.966	.964
	M $\downarrow$	.119	.027	.034	.024	0.27	.023	.021	.021	.018	.018	.016	.015
VT5000	S $\uparrow$	.679	.867	.864	.866	.862	.882	.871	.894	.894	.911	.898	.916
	F β $\uparrow$	.662	.872	.864	.857	.856	.888	.870	.895	.896	.915	.897	.917
	E $\downarrow$ $\uparrow$	.795	.899	.891	.907	.889	.924	.925	.930	.937	.945	.941	.951
	M $\downarrow$	.114	.043	.048	.042	.043	.035	.035	.033	.030	.026	.028	.023
Params(M)		–	79.8	83.1	4.6	–	69.9	24.1	40.0	28.7	198.8	93.8	95.9
FLOPs(G)		–	121.5	247.2	4.2	–	347.8	207.2	39.7	17.1	124.7	63.9	51.1

Evaluation Metrics. Precision–recall [61] represents the curves of precision and recall at different thresholds, where precision measures the accuracy of detection and recall measures the coverage of ground truth. S-measure [62] evaluates the structural similarity between the predicted map and the ground truth, which considers both the object region and the object structure. F-measure [63] combines precision and recall using a weighted average, and this study uses max F-measure as the evaluation criterion. E-measure [64] simultaneously evaluates global information and local pixel-wise matching information between the predicted map and the ground truth. MAE [65] calculates the mean absolute difference between the pixel values of the prediction map and the ground truth, which is one of the most critical evaluation metrics in SOD. To evaluate the performance of our method, we use the evaluation toolkit consistent with SLSOD [34] to compute these metrics.

Implementation Details. Our experiments are conducted on an NVIDIA 3090 GPU. We resize the input to 384×384 for depth, thermal, and RGB images. Single-channel depth images are repeated as three-channel inputs. Thermal images are used as three-channel data inputs. We employed various data augmentation techniques to avoid overfitting, including random flips, rotations, and crops. The P2T backbone is pre-trained on ImageNet, and the rest of the model's parameters are initialized according to PyTorch. We trained the model using the Adam [66] optimizer with a batch size of 3. The initial learning rate is $5e-5$ and decreases by a factor of 10 for every 100 epochs. We employ mixed precision training to speed up the training process and reduce resource consumption. The model convergence is within 200 epochs, which takes approximately 15 h.

4.2. Comparisons with SOTAs

4.2.1. Comparison with state-of-the-art RGB-D methods

For the comparison of RGB-D salient object detection methods, we select state-of-the-art methods from recent years, including D3Net [54], HAINet [17], VST [13], MMNet [67], EBMG [68], TriTransNet [12], DSA2F [69], C2DFNet [33], DigrNet [70], SPSN [71], RFNet [72], SLSOD [34], SwinNet [9], HINet [32], CAVER [11], and CATNet [14] totaling 16 methods. Among these methods, VST, EBMG, SPSN, TriTransNet, SwinNet, CAVER, and CATNet are based on transformers. It is worth noting that CAVER uses a ResNet101 backbone, and SLSOD adopts ImageNet pre-trained parameters. All saliency maps in the test are taken from the method author's open-source library and are evaluated using the same evaluation code.

Quantitative Evaluation. As shown in Fig. 5, the curve position corresponding to our method is higher, which indicates better performance of the proposed model. Table 1 demonstrates that our method significantly improves MAE and other metrics across the six tested datasets compared to state-of-the-art methods. In addition, our method has fewer parameters and FLOPs than other methods with similar performance. This advantage is attributed to our use of a smaller number of feature channels compared to other methods. At the same time, SAFM and AMFM employ a multi-head mechanism to enhance the utilization of feature channel information without further expanding the number of feature channels.

Qualitative Evaluation. The qualitative evaluation of the image quality is shown in Fig. 6. Rows 1~4 are fine-grained segmentation scenarios, and our method outperforms the others in the fine-grained

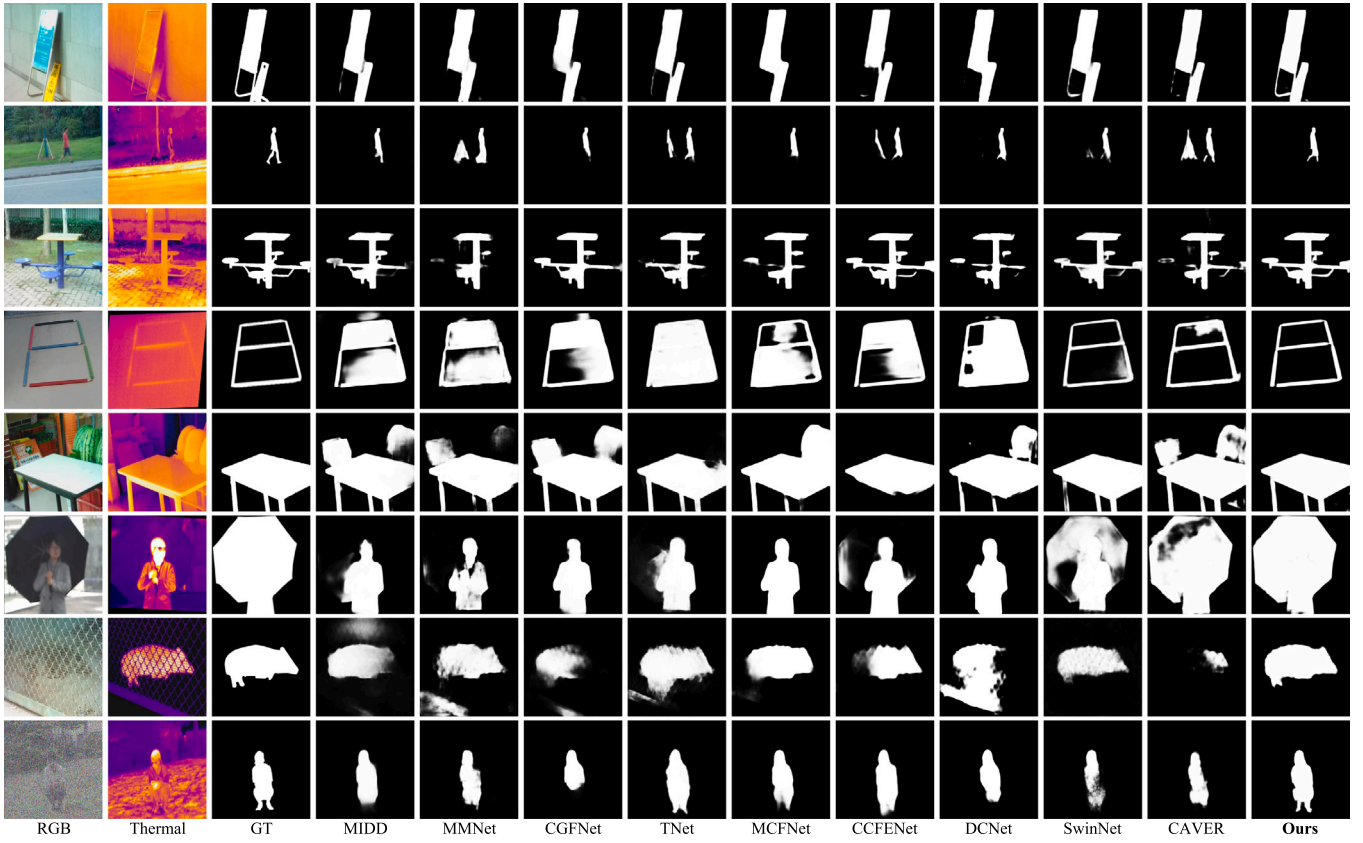


Fig. 8. Visual comparison of our model with other SOTA RGB-T models.

Table 3

Ablation analysis of our baseline gradually incorporating the proposed new components. The best scores are **bold**.

No.	Baseline	SAFM	AMFM	FMM	Params (M)	FLOPs (G)	STERE				SIP				VT5000			
							S $\uparrow$	F β $\uparrow$	E $\downarrow$ $\uparrow$	M $\downarrow$	S $\uparrow$	F β $\uparrow$	E $\downarrow$ $\uparrow$	M $\downarrow$	S $\uparrow$	F β $\uparrow$	E $\downarrow$ $\uparrow$	M $\downarrow$
1	✓				73.7	37.2	.927	.931	.942	.031	.914	.938	.944	.036	.911	.907	.938	.028
2	✓	✓			89.6	44.8	.927	.933	.957	.027	.911	.937	.945	.035	.912	.911	.947	.025
3	✓		✓		83.8	45.6	.928	.933	.953	.027	.912	.940	.944	.035	.914	.912	.949	.024
4	✓			✓	78.6	40.4	.928	.933	.957	.027	.915	.940	.950	.032	.914	.915	.950	.024
5	✓	✓	✓	✓	95.9	51.1	.932	.940	.962	.025	.917	.942	.953	.031	.916	.917	.951	.023

parts, such as wheel spokes and butterfly tentacles. Rows 4~5 are low-quality depth map scenes where our method outperforms the others regarding imaging edges and completeness. In rows 7~8, which are multi-target high-resolution scenes, our method identifies all targets completely and distinguishes people from shadows.

4.2.2. Comparison with state-of-the-art RGB-T methods

For the RGB-T salient object detection methods, we also select state-of-the-art models published in recent years, including MTMR [58], MIDD [43], MMNet [67], CGFNet [73], CSRNet [74], ADF [60], MCFNet [75], TNet [44], CCFNet [45], DCNet [76], SwinNet [9], and CAVER [11]. Among them, MTMR is a traditional method, while the others are deep learning-based methods.

Quantitative Evaluation. As shown in Fig. 7, the PR curves of our method on RGB-T datasets show powerful competitiveness compared with other methods. The quantitative comparison results of our method on the RGB-T datasets are shown in Table 2. Our method achieves an MAE of 0.023 on the VT-5000 dataset, which exceeds existing state-of-the-art methods by a large margin. In addition, these data illustrate that our PATNet has fine generalization ability on RGB-T datasets.

Qualitative Evaluation. The qualitative evaluation of RGB-T is shown in Fig. 8. Rows 1~3 are tiny target and fine-grained scenarios in RGB-T datasets, and it is evident that our method is better than others in terms of integrity and fine-grained details. Rows 4~6 are scenes

where the thermal image generates interference, and our method can balance these interferences to segment the salient targets accurately. Thanks to the multimodal synthesis capability of SAFM, our method performs better than the other methods in the scene shown in rows 7~8, where the lower quality of the RGB image requires the assistance of the thermal image.

4.3. Ablation study

In this section, we perform ablation experiments on all the proposed components. In the baseline, the P2T-base is used as an encoder, and the FPN structure is composed of convolutions as a decoder. The datasets involved in the ablation experiments are mainly the three with the largest number of samples in RGB-D and RGB-T, i.e., STERE, SIP, and VT-5000.

Effectiveness of All Modules. To verify the effectiveness of our proposed modules, we incrementally extend SAFM, AMFM, and FMM in the baseline. As shown in Table 3, No. 2~4 demonstrates that SAFM, AMFM, and FMM effectively enhance the baseline performance on the three datasets, respectively. The results of combining all the modules on the baseline are shown in No. 5, and the best results on all the datasets are achieved. We perform the same ablation experiments on VT-5000. As shown in the last four columns, the three components also gradually provide performance improvements. Compared to the baseline, our

Table 4

Effectiveness analysis of SAFM, AMFM, and FMM. SwinNet denotes the replacement of our module with the modules proposed in SwinNet. The best scores are **bold**.

Variant	STERE				SIP			
	S $\uparrow$	F $_{\beta}$ $\uparrow$	E $_c$ $\uparrow$	M $\downarrow$	S $\uparrow$	F $_{\beta}$ $\uparrow$	E $_c$ $\uparrow$	M $\downarrow$
SwinNet	.930	.934	.957	.028	.912	.939	.945	.036
Ours	.932	.940	.962	.025	.917	.942	.953	.031

Table 5

Effectiveness analysis of different attention operations. The best results are **bold**.

No.	STERE				SIP			
	S $\uparrow$	F $_{\beta}$ $\uparrow$	E $_c$ $\uparrow$	M $\downarrow$	S $\uparrow$	F $_{\beta}$ $\uparrow$	E $_c$ $\uparrow$	M $\downarrow$
1	.925	.929	.954	.029	.912	.935	.945	.034
2	.930	.935	.959	.027	.909	.936	.944	.035
3	.929	.936	.956	.027	.914	.939	.948	.033
4	.929	.934	.958	.027	.916	.942	.950	.033
5	.932	.940	.962	.025	.917	.942	.953	.031

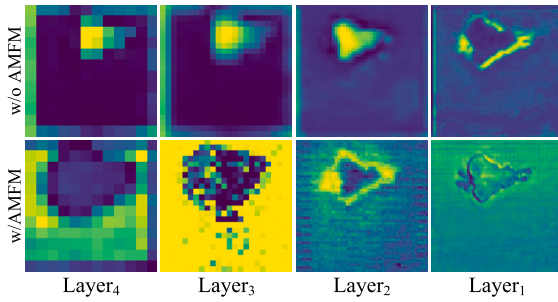


Fig. 9. Visual ablation experiments of AMFM. Layer₄ represents the output of the smallest scale feature map. Layer₁₋₃ are the intermediate feature maps obtained from the three AMFMs and the corresponding bilinear interpolation, respectively. The top row shows the results obtained using bilinear interpolation, while the bottom row shows the results obtained in AMFM before fusion.

proposed modules have increased parameters and FLOPs due to many linear projections and matrix multiplications in the attention modules. However, these costs are still within a reasonable range, and it may be possible to optimize the parameters and costs of these operations in future work.

In addition, we also replace our proposed modules with three modules used in SwinNet in the ablation experiments, including the spatial alignment and channel recalibration module, the edge-aware module, and the edge-guided decoder. In Table 4, it is apparent that our approach significantly improves over the modules proposed in SwinNet.

Effectiveness of Different Attention Operations. In order to validate the effectiveness of the attention operations proposed in SAFM and AMFM, we replaced the original modules with conventional self-attention or cross-attention. As illustrated in Table 5, No. 1 indicates the replacement of cross-scale attention in AMFM with self-attention, where low-resolution features are upsampled by bilinear interpolation and added with high-resolution features. In No. 2, cross-scale attention is substituted with cross-attention, where the input Key and Value come from interpolated low-resolution features. No. 3 denotes that the shared attention of SAFM is substituted with two independently performed cross-attention operations. In No. 4, the shared attention of SAFM is replaced by cross-attention with a single branching, and depth is exclusively employed as the Query. No. 5 indicates that both AMFM and SAFM utilize default settings. The results show that better performance can be achieved using shared attention in SAFM and cross-scale attention in AMFM. In contrast, all other attention manipulation strategies lead to some degree of performance degradation.

Effectiveness of Micro Design. As shown in Table 6, *w/o* multi-head means that the number of multi-heads in all SAFMs and AMFMs is set to 1, which is much less effective than using multi-head semantic information. “*w/o* PReLU” denotes the replacement of all PReLU in our

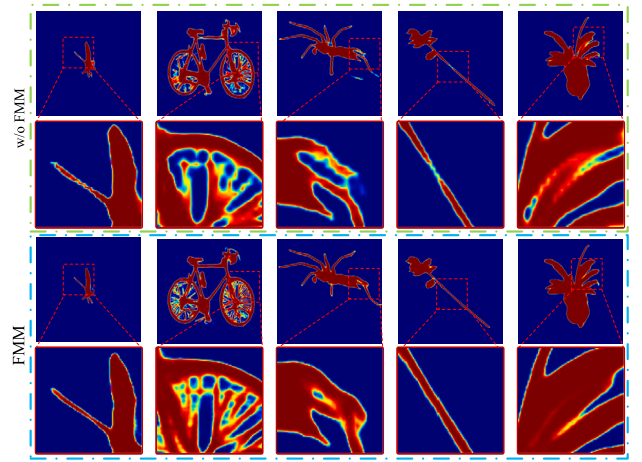


Fig. 10. Visual ablation experiments of FMM. The top two rows depict saliency maps obtained using the bilinear interpolation method, while the bottom two rows display the results obtained using FMM. The red and blue colors in the figure represent the foreground and background, while other colors indicate uncertain areas.

Table 6

Effectiveness analysis of Micro design. The best scores are **bold** and “*w/* $\star$ ” and “*w/o* $\star$ ” mean with/without $\star$.

Variant	STERE				SIP			
	S $\uparrow$	F $_{\beta}$ $\uparrow$	E $_c$ $\uparrow$	M $\downarrow$	S $\uparrow$	F $_{\beta}$ $\uparrow$	E $_c$ $\uparrow$	M $\downarrow$
<i>w/o</i> Multi-head	.929	.935	.958	.027	.908	.932	.943	.035
<i>w/o</i> PReLU	.929	.935	.955	.027	.913	.937	.947	.034
<i>w/o</i> Mul	.928	.933	.957	.028	.912	.939	.947	.033
<i>w/</i> all	.932	.940	.962	.025	.917	.942	.953	.031

method with the ReLU activation function. “*w/o* Mul” represents the replacement of dot product fusion with summation fusion in AMFM, resulting in a decline in model performance.

Visual Ablation Experiments on AMFM. We conduct ablation experiments on AMFM with visual comparisons. As shown in Fig. 9, these images show a butterfly with detailed antennae, and the top and bottom rows illustrate the intermediate feature maps obtained using different upsampling methods during the baseline processing of this image. Comparing the feature maps obtained using bilinear interpolation and upsampling with relational attention matrix modeling in AMAF, it can be observed from Layer₁ and Layer₂ that our method captures more detailed information in the shallow features.

Ablation Experiments on FMM. FMM is designed to map patches representing features to pixels to produce a detailed saliency map. To evaluate the performance of FMM on fine-grained images, we contrasted the fine-grained disparities between the saliency map produced by bilinear interpolation and that produced by FMM, as shown in Fig. 10. It can be seen that the saliency map generated by FMM produces less jaggedness with better continuity in the slender portion and higher confidence in the target edge.

In addition, we performed ablation experiments for channel weighting in FMM, as shown in Table 7. *w/o* CW means channel weighting is not used in FMM, and *w/CA* means CW is replaced with channel attention. The results show the positive effect of assigning channel weights via learnable parameters in the mapping. Moreover, using learnable parameters directly is more effective than channel attention, which requires additional feature compression.

4.4. Analysis of typical failure cases

Although the proposed PATNet exhibited a competitive performance, it cannot sufficiently address some complex scenarios, as shown in Fig. 11. In the first scene with a hollow structured object, our approach focuses on separating it internally from the background without

Table 7

Effectiveness analysis of Channel Weighting. The best scores are **bold** and “w/ ★” and “w/o ★” mean with/without ★.

Variant	STERE				SIP			
	S ↑	F _β ↑	E _g ↑	M ↓	S ↑	F _β ↑	E _g ↑	M ↓
w/CA	.930	.935	.958	.026	.913	.940	.947	.034
w/o CW	.931	.937	.959	.026	.916	.941	.950	.032
w/CW	.932	.940	.962	.025	.917	.942	.953	.031

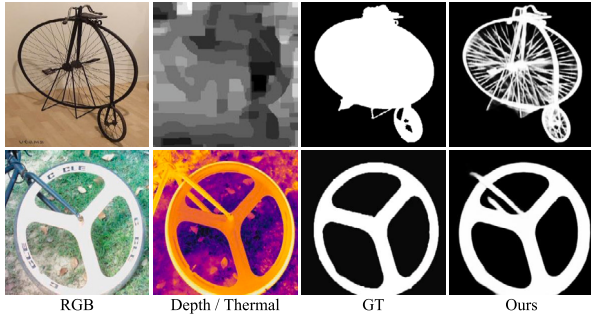


Fig. 11. Some failure cases of our method.

segmenting the object as a whole. The overlapping scene in the second scenario is also tricky, and our method incorrectly treats non-significant objects overlapping on the target as salient. For such problems, added capabilities will allow the model to decide whether to segment the entire region of the salient object by adjusting the prompt or post-processing according to different needs. In addition, increasing the number of samples of overlapping scenes in the multimodal SOD training set or improving the semantic capability of the model would be expected to solve the challenge of overlapping object segmentation.

5. Conclusion and future works

This study proposes PATNet to address the completeness and fine-grained issues in RGB-D and RGB-T multimodal salient object detection. To precisely locate saliency targets before the patch-level stage, P2T was used to extract multimodal features. A shared attention fusion module was proposed at the patch-level stage to increase the complementarity and consistency of RGB and other modal images. The proposed adjacent modeling fusion module further reduces the aliasing effect and enhances the cross-scale interaction capability. At the pixel-level stage, a fine-grained mapping module is proposed to map patches onto pixels for more accurate segmentation. The features are processed with different granularities in two stages to obtain a complete fine-grained significance map. In exhaustive experiments, PATNet outperformed 24 state-of-the-art methods on six RGB-D and three RGB-T datasets.

Although PATNet has shown promising results in multimodal salient object detection, there is room for improvement in the future. For example, the computational cost of attention in SAFM and AMFM can be reduced. Besides, more efficient and direct mapping in FMM is also worth exploring. In addition, hollow and overlapping scenes in multimodal SOD are challenges to be addressed.

CRediT authorship contribution statement

Mingfeng Jiang: Writing – review & editing, Writing – original draft, Supervision, Resources, Funding acquisition, Conceptualization. **Jianhua Ma:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiatong Chen:** Writing – review & editing, Writing – original draft, Visualization. **Yaming Wang:** Writing – review & editing, Writing –

original draft, Supervision, Funding acquisition. **Xian Fang:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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